# Offline Filipino Speech on iOS with Quantized VoxCPM2

**Ona Pacholo Asis**
QnSub Software Solutions

**Abstract** QnVoice bundles quantized VoxCPM2 for Filipino text-to-speech on iOS. In a matched 20-prompt desktop comparison with the app's earlier SpeechT5 system, Whisper's character error rate fell from 15.40% to 8.54%, while word error stayed above 50% for both systems. On an iPhone 15 Pro, median time to playback readiness for five warm requests fell from 7.87 s to 2.48 s after changes to the vocoder and reference encoding. The optimized runtime took 13.63 s on the first request after the synthesis session was released. In a separate Mac comparison, the vocoder change preserved the sample count for each of six paired renders, with a maximum per-clip root mean square difference of $1.36 \times 10^{-5}$ of full scale. Perceived naturalness remains a question for Filipino listeners.

*Keywords* Filipino, text-to-speech, mobile inference, voice conversion

## 1 Introduction

A user can type "kamusta dyan, san tayo kakain." in QnVoice. The app keeps that spelling on screen and sends "kumusta diyan, saan tayo kakain." to the synthesizer. When the same words come from a recording, the app changes the voice using the recorded audio. This distinction matters because a transcription mistake in the second case could replace words the user actually said [1].

Previous work on the Philippine Languages Database describes corpora for Filipino and other Philippine languages and their use in speech synthesis and voice conversion [2]. VoxCPM2 supports Tagalog and generates speech conditioned on an audio reference [3]. MobileSpeech is an example of a separate approach developed for mobile zero-shot synthesis [4]. We compare QnVoice's earlier and newer speech systems on matched desktop prompts and examine how two runtime changes affect playback on an iPhone.

## 2 Implementation

QnVoice uses common words and language detection to route typed Filipino to the synthesizer. Free-form English uses an int8 ONNX export of Helsinki-NLP's Marian English-to-Tagalog model [5]. Build 12 looks up the bundled Filipino wording for an exact match to a displayed localized sample. A reviewed translation also covers the English "self-destruct" prompt and its numeric variant [1].

Text preprocessing standardizes curly quotes, guillemets, and full-width punctuation. It replaces dash punctuation with spaces and rewrites peso signs and leading minus signs as words. Input with no letter or number is rejected after normalization. The synthesizer expands "san" before words such as "tayo" and preserves names such as "San Juan." The app displays the original typed text [1].

VoxCPM2 ships as two GGUF files totaling 2.080 GB (2,080,115,072 bytes). The BaseLM uses Q4_K_M. The acoustic file uses Q8_0 for matrices in ResidualLM, LocEnc, and LocDiT. The vocoder and stop predictor retain their source precision. The 4.48 s synthetic reference is mono audio at 16 kHz. QnVoice uses the pretrained weights [1, 3].

The runtime uses a 1,024-token context and writes mono audio at 48 kHz. It vocodes up to four latent patches per pass and carries state between passes. The optimized AudioVAE implements transposed convolution as a matrix multiply followed by overlap-add where the kernel width is twice the stride. The first render encodes the synthetic reference, and subsequent renders reuse those features for the lifetime of the synthesis session [1].

The app publishes audio once a phrase completes. At load time it requires at least 5.5 billion bytes of physical memory and 2.6 billion bytes of available process memory. It releases an idle synthesis session after 20 s [1].

For Filipino recordings without transcript edits, OpenVoice V2 runs through Core ML. The app mixes high-frequency detail from the source back into the converted audio. Whisper transcribes both waveforms. If the detected language or normalized words disagree, QnVoice falls back to the source recording and lowers its pitch toward the reference where possible. The two transcripts can still agree when the recognizer makes the same error on both clips [1, 6, 7].

## 3 Methods

### 3.1 Desktop Comparison

The two desktop systems rendered the same 20 project-authored prompts. The VoxCPM2 harness called the native runtime directly, bypassing the app’s spelling rules. It generated 48 kHz audio with the earlier voice reference used in the build 5 period. SpeechT5 used digest-checked assets and its historical ONNX Runtime decoding loop to produce 16 kHz output [1, 8].

Whisper Base ran on the CPU with Tagalog forced, temperature set to zero, and conditioning on previous text disabled. The scorer case-folded the transcripts, stripped diacritics except the tilde in ñ, and extracted tokens matching [a-zñ0-9]+. Word error rate (WER) used those tokens. Character error rate (CER) used the same text with tokens joined. Both metrics divide the total Levenshtein edit count by the total number of reference units.

### 3.2 iPhone Playback and Regression Tests

The paired playback benchmark ran on an iPhone 15 Pro with iOS 27.0 in a test project stamped build 10. It released the synthesis session before sending the same six typed Filipino requests in a fixed order. The earlier and optimized configurations each ran once. The five requests after the first are labeled warm. The test required a new take for every request, including the nonconsecutive repeat of “Salamat po.” [1].

Playback readiness was timed from the request until rendering finished and the player reported playback. A second measurement ended when the app’s playback level meter reached 0.2 on its own scale. The optimized run included two extra requests containing em dashes, one Filipino and one English. We report these separately from the matched prompts [1].

Earlier phone tests used bundled Whisper Small Q5_1 to transcribe a five-word phrase, allowing specified spelling variants. The 2.40 s recording fixture entered through the capture callback. Pitch was estimated by normalized autocorrelation on 40 ms frames with a 10 ms hop [1].

### 3.3 Vocoder Comparison

On a Mac, we used Metal to render the same six requests with each vocoder. Both configurations used the same weights, decoding settings, and four-patch batches, and encoded the reference on each request. We timed renders after model load and compared paired 48 kHz waveforms by sample count and root mean square difference [1].

### 3.4 Text Input Checks

Build 11 device checks covered preset playback and dash-only input. The build 12 Mac audit ran the app's input normalization and translation tokenizer on 119 English catalog entries and 94 symbol probes embedded in a sentence. A separate check used 969 sample–locale pairs to verify that localized text resolved to the bundled Filipino wording and unchanged source text tokenized. These checks assess input handling and routing. Five selected build 12 regressions ran on the iPhone 15 Pro with iOS 27.0 [1].

## 4 Results

### 4.1 Desktop Transcription

VoxCPM2 reduced character edits on 13 of the 20 prompts, with six increases and one tie. Word edits fell on nine prompts, rose on eight, and tied on three. The aggregate totals show a larger difference at the character level (Table 1) [1].

**Table 1. Whisper errors on 20 matched development prompts**

| Synthesis pipeline | Character edits | CER | Word edits | WER |
|---|---|---|---|---|
| Quantized VoxCPM2 | 56 / 656 | 8.54% | 68 / 130 | 52.31% |
| Historical Filipino SpeechT5 | 101 / 656 | 15.40% | 70 / 130 | 53.85% |

The denominators are reference characters and words.

Consider "Hindi. Bukas na lang tayo magkita." Whisper wrote "Hindi, bukas nalang tayo mag kita." It joined "na lang" and split "magkita," producing four word edits among six reference tokens even though the letters matched exactly. The transcript alone cannot show whether the audio or the recognizer introduced those boundaries.

### 4.2 iPhone Playback

The median time to playback readiness for the five matched warm requests fell from 7.87 s to 2.48 s. In the optimized run, all five reached the playback meter threshold within 4.15 s. The first request after session release took 13.63 s to reach playback readiness (Table 2) [1].

**Table 2. Playback timing for matched typed requests on an iPhone 15 Pro**

| Request and endpoint | Earlier (s) | Optimized (s) |
|---|---|---|
| First request to playback readiness | 17.37 | 13.63 |
| Warm requests to readiness, median | 7.87 | 2.48 |
| Warm requests to readiness, range | 6.47–13.20 | 2.10–3.96 |
| Warm requests to meter threshold, median | 8.14 | 2.73 |
| Warm requests to meter threshold, range | 6.78–13.40 | 2.42–4.15 |

In the optimized run, “Salamat — po.” reached playback readiness in 2.23 s. “Thank you — for your help.” took 7.56 s through the path that includes translation. In a separate build 9 test, an immediate repeat of “Salamat po.” played from the waveform cache in 0.19 s, after a first request of 7.94 s [1].

The build 7 recording test retained the original audio file and produced a 2.40 s output matching the fixture duration. Whisper recovered all five expected words from the typed and recorded outputs after applying the specified spelling equivalences. Median voiced pitch was 327 Hz and 137 Hz for the build 6 and build 7 typed clips. The corresponding recording clips were 330 Hz and 127 Hz [1].

### 4.3 Vocoder Comparison

Median render time in the six paired Mac requests fell from 4.09 s to 1.90 s. The median time attributed to the vocoder fell from 2.54 s to 0.35 s. Every pair had the same sample count. Per-clip root mean square differences were $1.04 \times 10^{-5}$ to $1.36 \times 10^{-5}$ of full scale [1].

### 4.4 Text Input Checks

Build 11 device logs recorded playback of a translated channel greeting and the reviewed “self-destruct” preset, and rejection of dash-only input. Some test attempts failed before playback because a transcript destination folder was missing. On build 12, all five selected iPhone tests passed. The pasted-text test preserved the source text, retained the amount in the translation, and started Filipino playback [1].

The build 12 Mac audit tokenized all 119 English entries and all 94 selected symbol probes. All 969 sample–locale pairs passed across 51 locale identifiers. Of these, 893 matched a localized string and returned the expected Filipino wording. The 76 entries identical to their source strings passed tokenization. A pasted example with guillemets, a zero-width space, a full-width exclamation mark, and a peso amount produced the same tokens as its normalized English equivalent. The English tokenizer still rejected the tested emoji and the free-form Chinese and Russian examples [1].

## 5 Discussion

The historical desktop results show closer character agreement for VoxCPM2 under the same prompts and recognizer. The two synthesis systems used different speakers, decoders, and sample rates, so the comparison measures the systems as configured. The prompts came from development, and the phone transcription check covers a single colloquial expression. These results describe transcription agreement, while perceived naturalness requires listener ratings.

The phone benchmark used one fixed-order run per configuration on one device. The configurations differed in both vocoding and reference caching. The Mac probe isolated vocoding and found small numerical differences across the tested waveforms. For recorded speech, the logs sometimes leave the origin of the final audio uncertain because they omit whether the fallback ran.

Project notes report that the historical Mac test ran with network access denied. The phone tests used bundled models and a local recognizer. Repeated trials across phones could measure timing variation, peak memory, and thermal effects under network isolation [1].

A follow-up could compare verbatim and normalized speech for standard Filipino, informal spelling, and English–Tagalog code-switches [9]. Fluent listeners would transcribe clips and rate naturalness under blinded conditions with human readings as controls. Accepted spelling variants, recruitment, and listener-level uncertainty should be reported [10, 11].

## Data and Code Availability

The pinned Hugging Face release hosts the two GGUF weights, synthetic reference, and checksums [12]. The weights are available under Apache 2.0. QnSub Software Solutions permits personal and commercial use of the synthetic reference and speech it generates at no charge. An app that uses the reference for synthesis must cite this paper in its About screen, credits, or public documentation [12]. The GitHub companion project provides a verified downloader, model provenance, and the manuscript. GitHub companion project · Pinned Hugging Face release

The project records model provenance in Docs/filipino-model-manifest.json and retains its evaluation data in local build archives. Table 1 uses the native-bounded-eval and legacy-speecht5-eval data under build/model-tools/qnvoice-filipino/v2/. The paired vocoder waveforms and compare_waveforms.py are in build/filipino-five-seconds/.

Table 2 uses phone.log from build/filipino-warm-latency.IlLgjp/ and build/filipino-five-seconds/. The build 11 phone logs are in build/filipino-punctuation/. The build 12 audit harness, catalog snapshot, and phone results are in build/filipino-character-audit/. The recording fixture is Tests/Fixtures/filipino-colloquial.wav.

## References


[1] QnVoice. Filipino voice implementation, model manifest, regression tests, and local build archives. QnVoice 1.0.0, builds 5–12, 2026.

[2] Guevara, R. C. L., Cajote, R. D., Bayona, M. G. A. R., and Lucas, C. R. G. Philippine Languages Database: A Multilingual Speech Corpora for Developing Systems for Low-Resource Languages. SIGUL, pp. 264–271, 2024. ACL Anthology

[3] Zhou, Y., et al. VoxCPM2 Technical Report. arXiv:2606.06928, 2026. arXiv

[4] Ji, S., Jiang, Z., Wang, H., Zuo, J., and Zhao, Z. MobileSpeech: A Fast and High-Fidelity Framework for Mobile Zero-Shot Text-to-Speech. ACL, pp. 13588–13600, 2024. ACL Anthology

[5] Helsinki-NLP. OPUS-MT English to Tagalog (opus-mt-en-tl). Model card

[6] Qin, Z., Zhao, W., Yu, X., and Sun, X. OpenVoice: Versatile Instant Voice Cloning. arXiv:2312.01479, revised 2024. arXiv

[7] Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., and Sutskever, I. Robust Speech Recognition via Large-Scale Weak Supervision. ICML, pp. 28492–28518, 2023. PMLR

[8] internetoftim. speecht5_tts-pld-fil-ONNX. Model card

[9] Santa Maria, A., and Enghels, R. The Corpus of English–Tagalog Code-switching: an integrative corpus of linguistic contact effects. Scientific Data, 2026. DOI

[10] International Telecommunication Union. A method for subjective performance assessment of the quality of speech voice output devices. ITU-T Recommendation P.85, 1994. ITU

[11] Chiang, C.-H., Huang, W.-P., and Lee, H.-Y. Why We Should Report the Details in Subjective Evaluation of TTS More Rigorously. Interspeech, 2023. ISCA Archive

[12] Asis, O. P. QnVoice Filipino VoxCPM2 GGUF. QnSub Software Solutions, 2026. Hugging Face